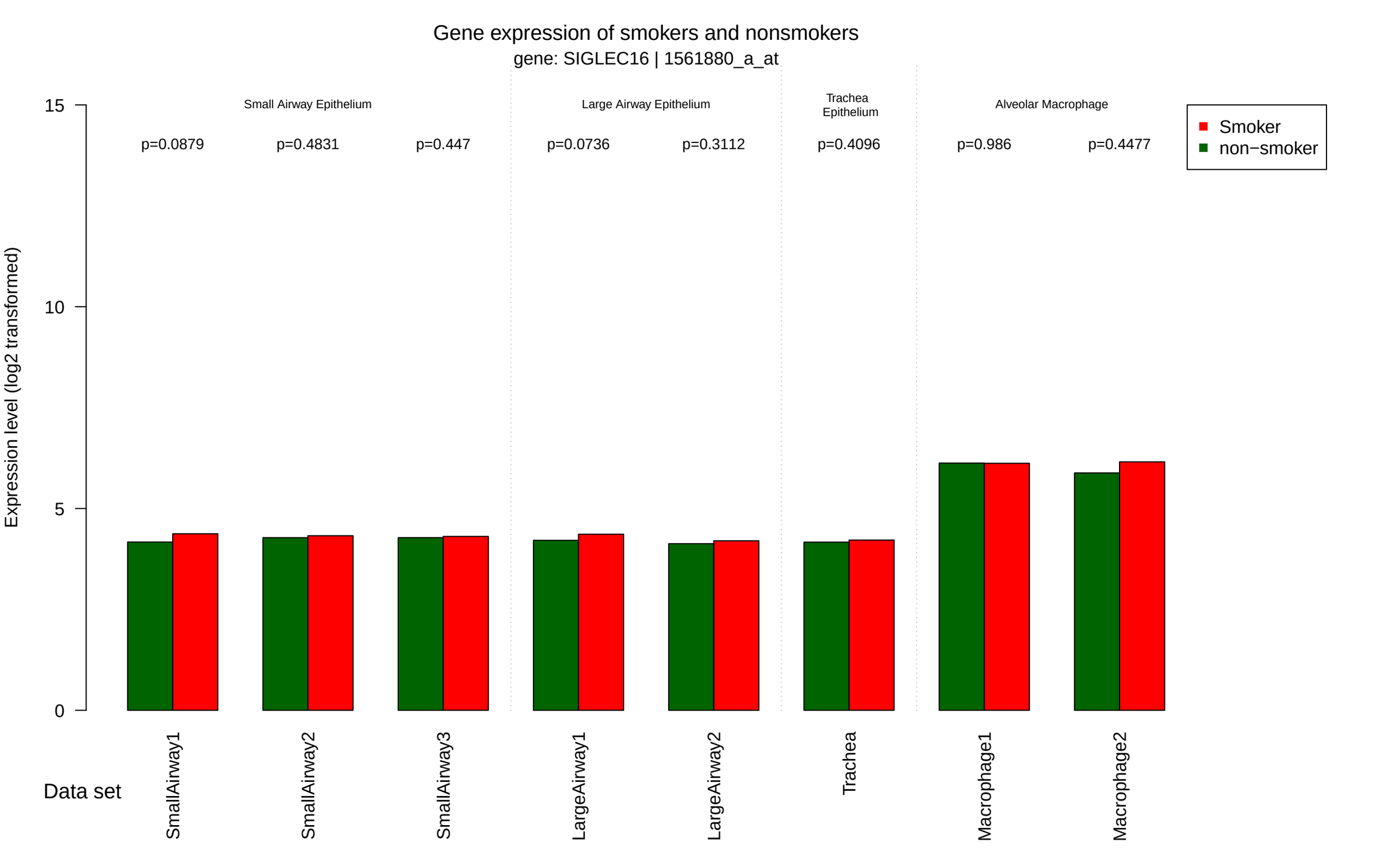
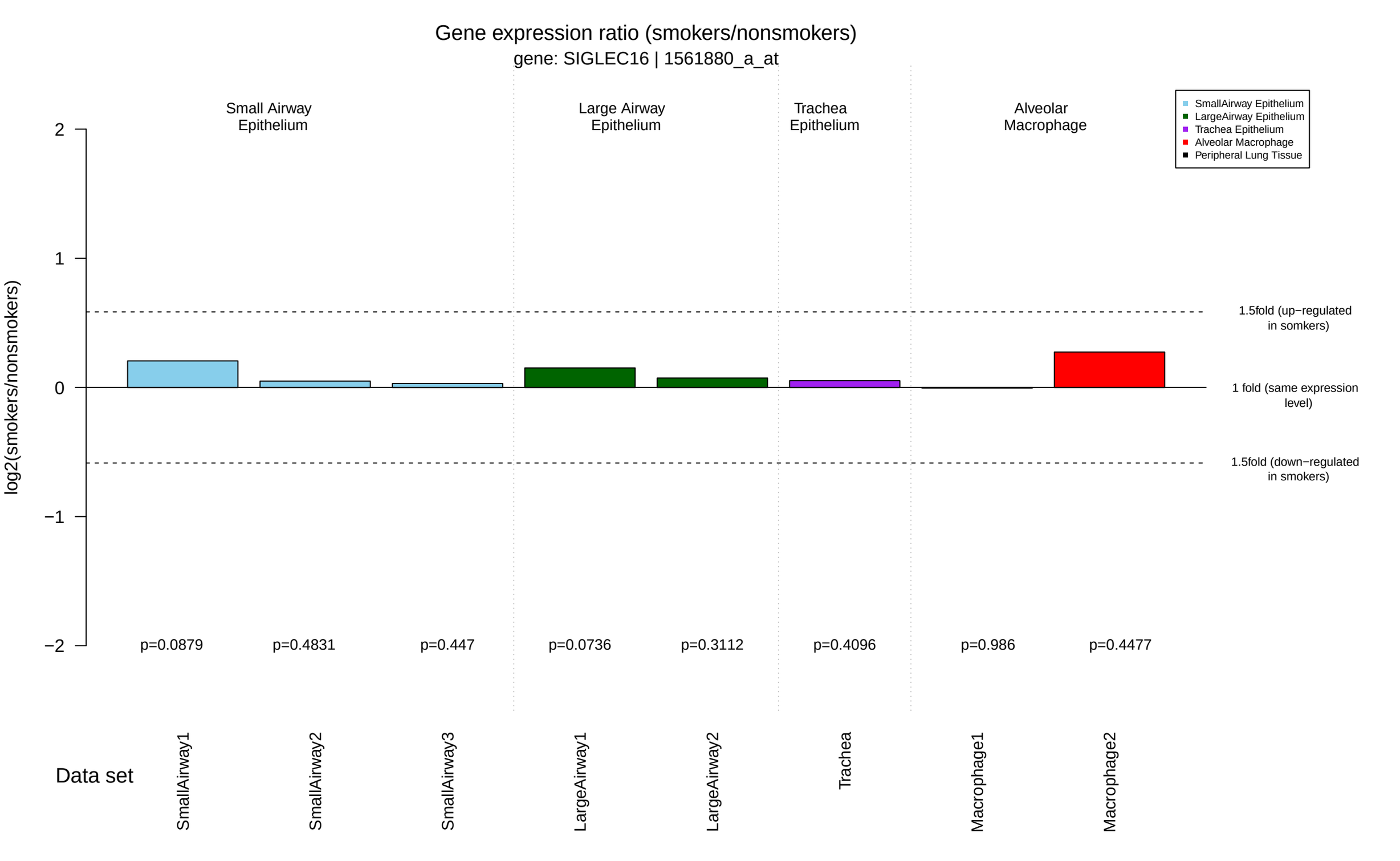
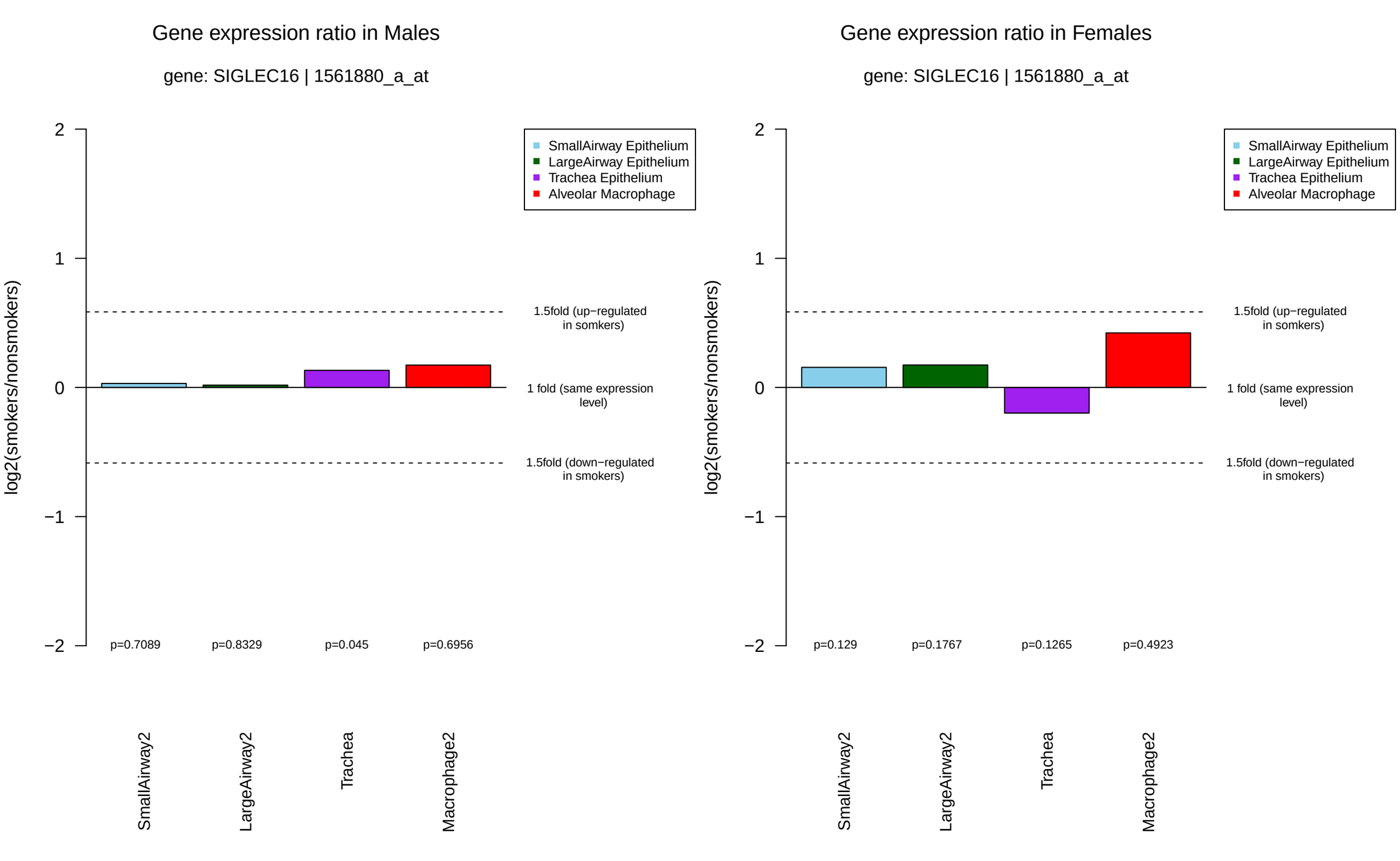
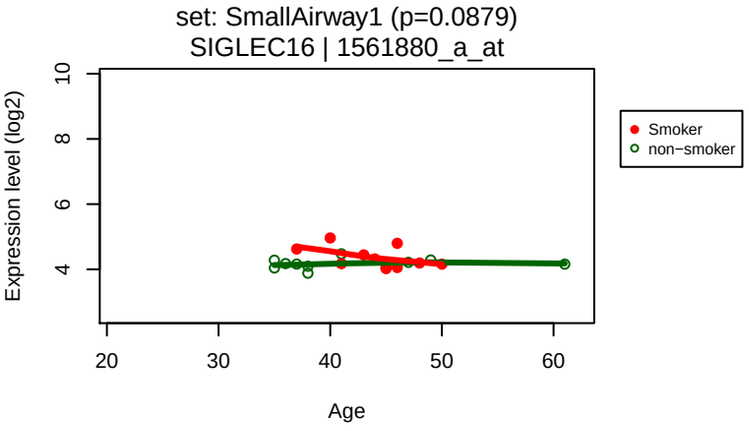
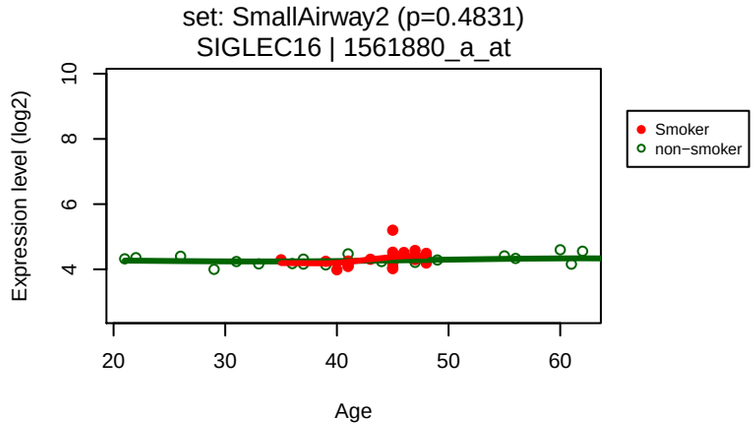
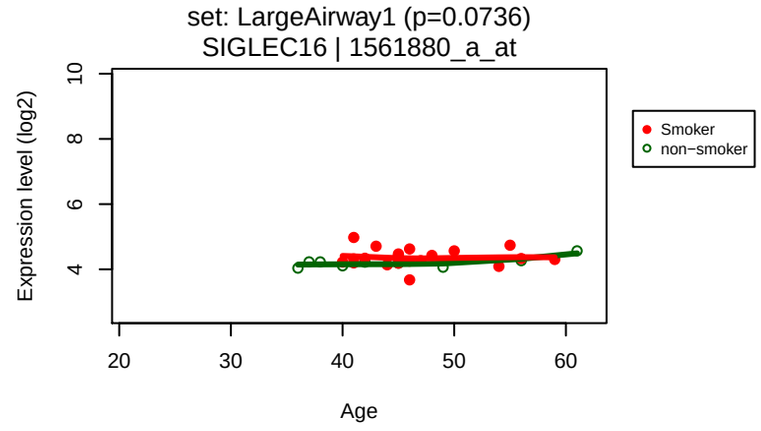

Gene expression of smokers and nonsmokers
gene: SIGLEC16 | 1561880_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0879
p=0.4831
p=0.447
p=0.0736
p=0.3112
p=0.4096
p=0.986
p=0.4477
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

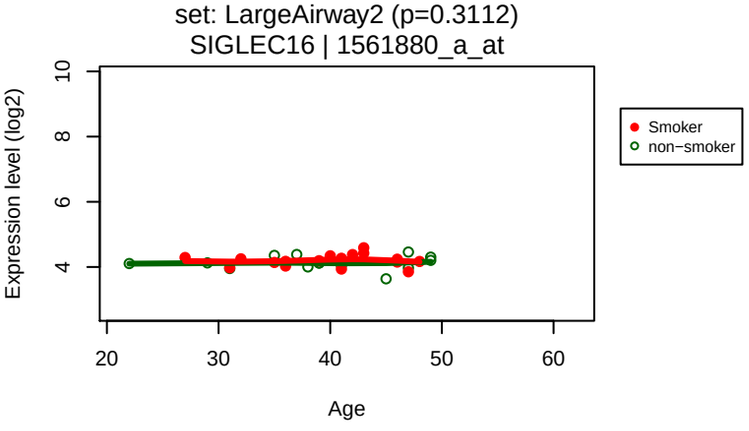

Gene expression ratio (smokers/nonsmokers)
gene: SIGLEC16 | 1561880_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0879
p=0.4831
p=0.447
p=0.0736
p=0.3112
p=0.4096
p=0.986
p=0.4477
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

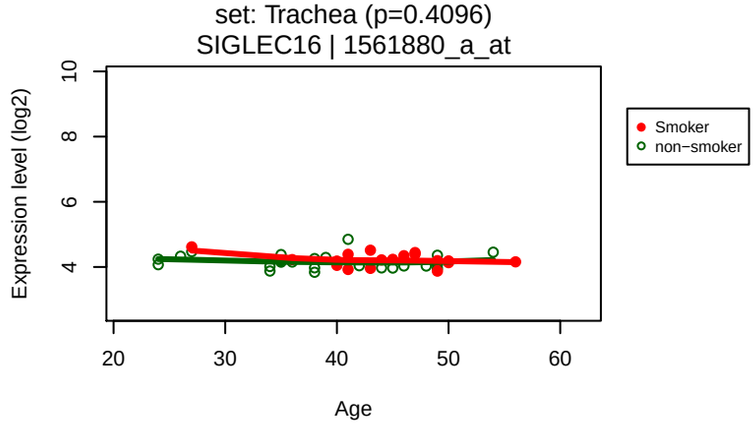

Gene expression ratio in Males
gene: SIGLEC16 | 1561880_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7089
p=0.8329
p=0.045
p=0.6956
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIGLEC16 | 1561880_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.129
p=0.1767
p=0.1265
p=0.4923
SmallAirway2
LargeAirway2
Trachea
Macrophage2

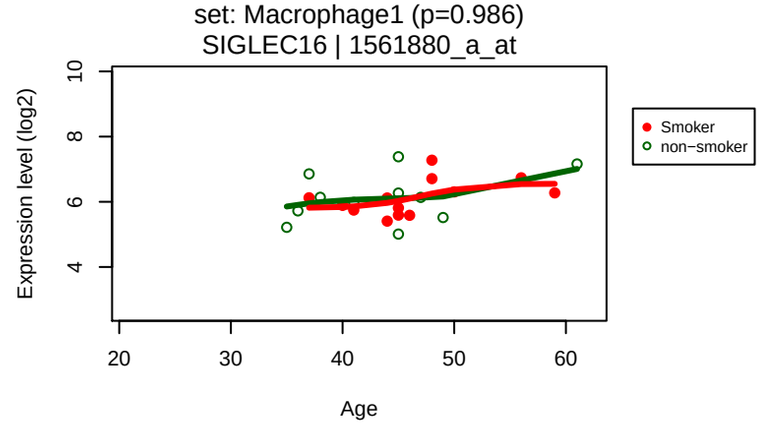

set: SmallAirway1 (p=0.0879)
SIGLEC16 | 1561880_a_at
Expression level (log2)
Age
Smoker
non−smoker

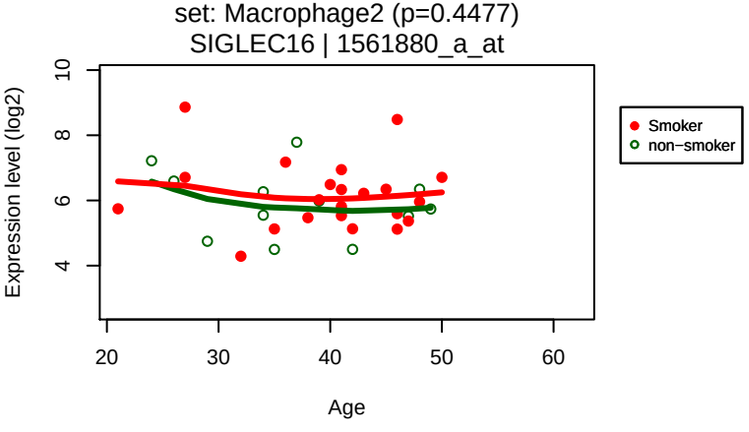

set: SmallAirway2 (p=0.4831)
SIGLEC16 | 1561880_a_at
Expression level (log2)
Age
Smoker
non−smoker

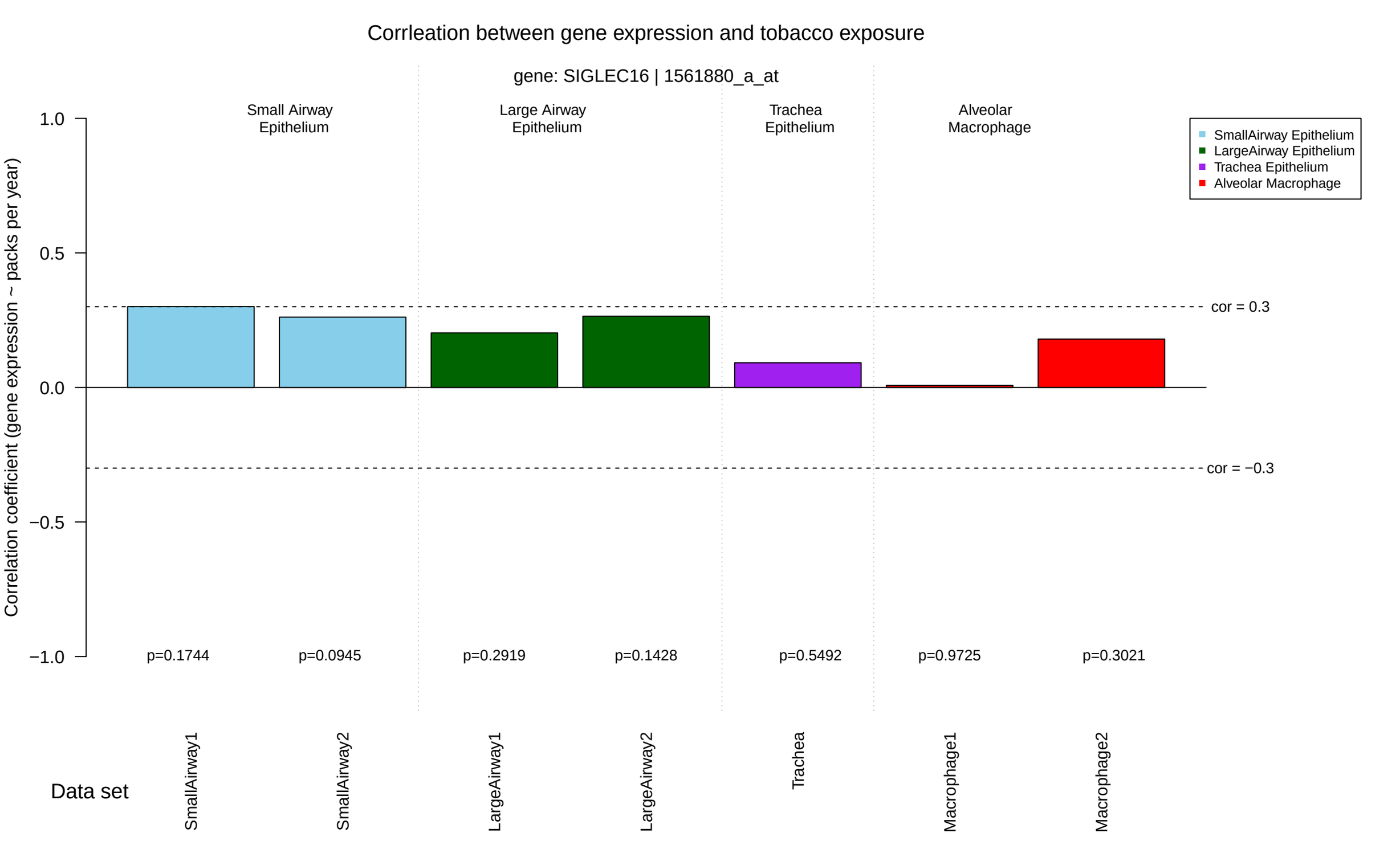

set: LargeAirway1 (p=0.0736)
SIGLEC16 | 1561880_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3112)
SIGLEC16 | 1561880_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4096)
SIGLEC16 | 1561880_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.986)
SIGLEC16 | 1561880_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4477)
SIGLEC16 | 1561880_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIGLEC16 | 1561880_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1744
p=0.0945
p=0.2919
p=0.1428
p=0.5492
p=0.9725
p=0.3021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2